

PEERACHET KHANITSON

AI/ML ENGINEER · DATA SCIENTIST · DATA ANALYST · AI DEVELOPER

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SUMMARY

Computer Engineering graduate with a strong focus on Artificial Intelligence and Machine Learning. Hands-on experience analyzing datasets and building predictive models for real-world scenarios. Motivated to turn data into actionable insights and develop intelligent solutions while growing as an AI Engineer.

EDUCATION

Mae Fah Luang University - Bachelor of Engineering - Computer Engineering | GPAX: 3.68

EXPERIENCE

Mitsubishi Electric Thai Auto-Parts Co., Ltd. Internship Jan - Apr 2026 (4 Months)

- Data Collection & System Development: Digitized operational workflows in a centralized Power Apps solution for standardized, structured data and reliable downstream analysis.
- Data Pipeline & Automation: Built automated data-sync workflows in Power Automate, improving data integrity by eliminating manual entry errors.
- Data Structuring & UI Design: Designed structured data schemas and intuitive UIs that boost operational efficiency and make end-user data capture more accurate.

PROJECTS

PM2.5 Level Classification (Senior Project) · GitHub

- Built an end-to-end multi-class ML model predicting PM2.5 in 5 levels per the Thai Air Quality Standard, using ~500,000 weather records from Kaggle.
- Cleaned and preprocessed data (missing-value imputation, trigonometric wind encoding) and fixed severe class imbalance by comparing oversampling methods, including SMOTE variants.
- Compared 8 ML/DL models; Extra Trees with Borderline-SMOTE performed best: 92.7% accuracy/F1-score and 99.3% ROC-AUC, evaluated with a confusion matrix.

Credit Scoring Model (Logistic Regression) · GitHub

- Built a logistic regression model to assess loan default risk.
- Optimized features (WOE, IV), resolved multicollinearity, validated with ROC-AUC and Gini.
- Created a score-grading framework with cut-off simulations to optimize loan approval and cut bad debt.

Computer Vision with OpenCV · Object Detection (GitHub) · Face Detection (GitHub)

- Built real-time video object detection with OpenCV DNN and a pre-trained MobileNet SSD (20 categories), filtering at 0.3 confidence and drawing boxes with labels and confidence scores.
- Wrote a Haar Cascade script to detect faces, eyes and cats in static images, with grayscale conversion, ROI extraction and bounding-box visualization.

TECHNICAL SKILLS

- **Languages:** Python, C, Java · **Databases:** MySQL
- **ML / DL / NLP:** Classification, Regression, Clustering, Ensemble (RF, ET, GB), Deep Learning (CNN, LSTM), NLP, Predictive Analytics
- **Computer Vision:** OpenCV, Object/Face Detection, Image Enhancement, Spatial/Frequency Filtering, Restoration, Color Processing
- **Data & Evaluation:** EDA, Data Cleaning, Feature Engineering, Data Transformation, Imbalanced-Data Handling, Cross-Validation, Bias/Variance, Accuracy/Precision/F1/ROC-AUC
- **Libraries:** Pandas, NumPy, Matplotlib, Seaborn, Scikit-Learn, TensorFlow/Keras
- **Tools & Platforms:** VS Code, Jupyter, Colab, Git/GitHub, Power Platform
- **Web & Mobile:** HTML, CSS, JavaScript, React, Flutter (Dart)

CERTIFICATE

- Machine Learning & Data Analytics: Predicting Consumer Behavior for Business Decisions
- Natural Language Processing (NLP): Solving Business Problems with NLP and Deep Learning
- Computer Vision & OpenCV: Object/Face Detection and Deep Neural Network Workshop
- Artificial Intelligence: In-Depth Neural Networks & AI